

TONIGHT

Numbers you can defend, charts that don't lie

Counting is done. Tonight the data starts answering: which number to quote, and how to show it so a manager can argue with it.

Data Analysis & Machine Learning for Business

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Everything we have, in the order we met it.

01

Notebook

code cell vs
markdown cell
run order, variables
int float str bool
`[...] {key: value}`

02

Our data

task, cells, items,
container
one row = one scan
nine columns
StartTask ...
FinishTask

03

pandas

select, filter, sort,
derive, aggregate
groupby: split, apply,
combine
a join glues two
tables on a shared
key
`how = left / inner
/ outer`
count rows before
and after

04

SQL

the same verbs,
where company data
lives
pull with SQL, polish
with pandas

05

The pickle

not a table, a saved
Python object
the pay rules live
here
**and there is no shift
column**

live, cell 0b

Counting is done. Tonight the data answers.

WEEKS 1 TO 2, DONE

Compute any number

Filter, group, join, and check the row count.

Level one of the ladder: what happened.



TONIGHT

Trust the number, then show it

Which numbers deserve trust, and how to present one so that someone else can check you.

Level one, done properly. The promise from week one.



FROM NEXT WEEK

Why, and what next

Real difference or luck, then models.

Levels two and three: why, and what will happen.

Ask Name one number from last week you would put in a report. Would you defend it to the manager?

Five steps, and then your homework.

01 One number always lies a little

mean vs median

02 Quantiles are promises

p50, p90, p99

03 The boxplot, taken apart

finally explained

04 Two warehouses, two verdicts

after the break

05 Charts that don't lie

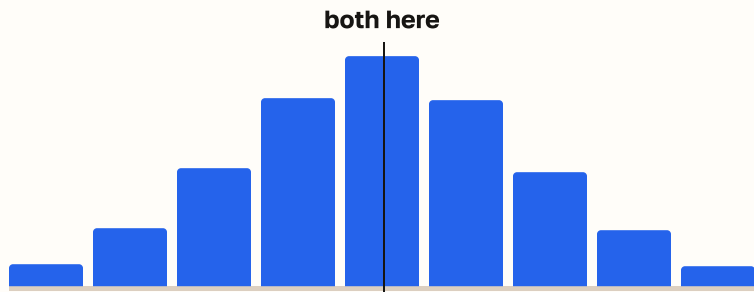
four crimes, ours

HW1 goes out at the end: **your own warehouse, 40% of the grade.**

One number always lies a little.

SYMMETRIC

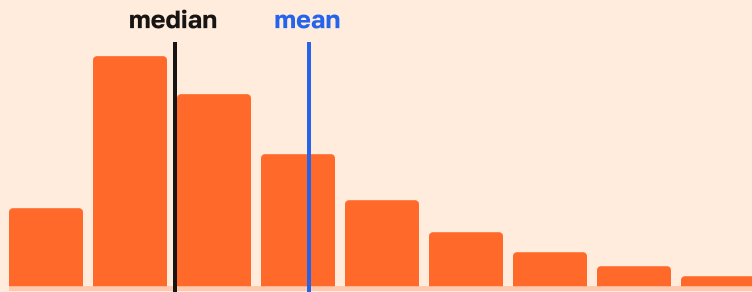
mean = median



Everyone is happy, and nothing tells you which case you are in except looking.

SKEWED, THE USUAL CASE

the tail drags the mean



Average salary sits above median salary, because oligarchs exist. Same arithmetic, different question.

Median answers what is typical. **Mean** answers what it sums to. Money likes means.

05 / 13

Hands up. In our warehouse, is the mean gap between two scans above or below the median?

Demo 01

Honest statistics on our own warehouse.

Every block asks a question, answers it, and then says plainly what the answer does **not** mean.

- 1 The per-task table, step by step, and the 160 unfinished tasks left out on purpose
- 2 How long does a task take? mean 16.9 against median 15.9 min
- 3 How fast do we scan? Gaps between scans, median 7.2 s against mean 13.7 s
- 4 **The useless histogram:** 95.4% of the gaps in one bar, then two repairs
- 5 Gaps become shifts, and the threshold is your judgement call
- 6 Quantiles, what a boxplot actually is, then seven sectors compared

Quantiles are promises. Averages are not.

p50

16 min

Half our tasks finish within sixteen minutes.

p90

28 min

Nine out of ten finish within twenty-eight.

p99

44 min

The worst one percent run past forty-four.

Clients, SLAs and boards live on promises. One bad day breaks a promise made with an average, so operations teams speak in quantiles instead.

I say: on average, sixteen minutes. **What is your follow-up?**

Meet BRAVO.

SITE ONE

ALPHA

The warehouse you already know. Week of 02 to 08 March.

SITE TWO, NEW

BRAVO

Same processes, same table schema, same week. A different site.

FROM THE DATA DICTIONARY Ids are local to a site. Join within a warehouse, never across.

Which warehouse is better?

Better by what? Faster, fewer errors, cheaper? Name your metrics now, before we compute anything. Most of them we will compute tonight, and they will disagree with each other.

Demo 02

Six operations metrics say BRAVO loses everything. Then we open the pickles and the money says otherwise.

THE FLIP

5.4 ₺ ALPHA **5.0 ₺** BRAVO

Pick pay per 1,000 ₺ of value moved. The worse warehouse moves a rouble cheaper.

One loader, two sites, and one task table built the same way on both

The scoreboard: wrong scans 18.6% against 26.8%, median 126 against 99 picks per hour

Slower, or just a lower average? Two histograms, then boxes

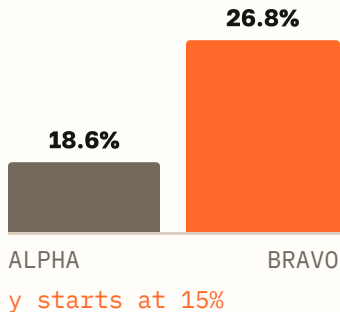
Money: 208M against 117M ₺ moved, and the pay rates differ, 7 ₺ against 6 ₺

The flip: pay per 1,000 ₺ moved, error rate against labour cost

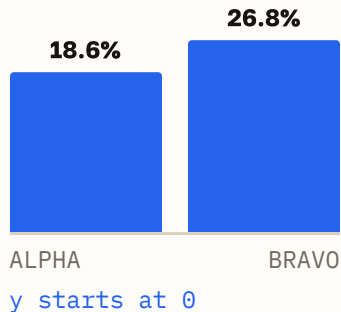
What do you tell the manager?

Same data, two emotions. The axis did it.

BRAVO is a catastrophe!!



Wrong-scan share by site, week of 02 to 08 March



- 01 Bars start at zero.
- 02 Show the spread, not only the mean. Two groups means two shapes.
- 03 The title says what is drawn. The conclusion goes underneath, in words.
- 04 Label the data, the axes and the units.
- 05 Cut something out of a chart? Say how much. Used three times already tonight.

A truncated axis is usually a default, not a lie. That is why the rule is mechanical.

Demo 03

You commit four chart crimes on purpose, so you recognize them in someone else's slide deck.

Crime 1	The truncated axis. 18.6% against 26.8%, drawn both ways
Crime 2	The arguing title. Both titles in the exhibit argued
Crime 3	Means without spread. Bars against boxes, and the 2.5% the boxes hid
Crime 4	4,077 dots in a cloud, then buckets with counts. Medians 73 to 143
Fix	One chart from default to report grade, conclusion printed underneath
AI	One prompt, an interactive plotly page, then verify its numbers

The bad charts stay on plain matplotlib defaults on purpose.

Your own warehouse. Four different businesses, not four sets of numbers.

Hypothesis in words **before** the code, plus what would change your mind. Start from `hw1_template.ipynb`.

Part 0 Meet your data: period, size, workers, data quality. Some columns are empty by design.

Part 1 A productivity metric, your definition, outliers handled consciously.

Part 2 Three hypotheses: task size, item properties (a join), and one of your own.

Part 3 **The fork: earnings per shift.** The pay rules are in the pickle, “shift” is not. Define it, defend it.

Part 4 Ten sentences for a human: what you found, what you would recommend, what you would check next.

YOUR SITE, BY DM TONIGHT

Podolsk / Kazan / Ekaterinburg / Novosibirsk

The whole folder as it is. Nothing to cut.

RUBRIC

all four: **up to 8 of 10**

40

correct

25

logic

20

business

15

artifact

THE LAST TWO POINTS, EXACTLY ONE

a second own hypothesis on a new dimension, an AI html one-pager for the manager, or a Session 4 test read honestly

Due **Sep 30, 18:00**, notebook and html by DM | Warehouse name in the first cell |

Late: 10% of the earned score per started day, max three

Six ideas to keep.

01 Distribution first, number second.

02 Median is typical, mean is totals. Skew splits them.

03 Quantiles are promises: p50, p90, p99.

04 A boxplot is q1, median, q3 with 1.5 IQR whiskers. A dot past the whisker is data, not an error.

05 Metric choice is a decision. Two honest metrics, two verdicts.

06 Bars from zero, spread on show, title says what is drawn, too many dots become buckets with counts. And say how much you cut.

NEXT WEDNESDAY

Sectors looked different. BRAVO looked worse. How do we tell a real difference from luck? Hypothesis testing: p-values with pictures, no formulas, and the tools to defend every claim in your HW1.